

Yoo Jin (Sharon) Cha

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RESEARCH INTERESTS

Numerical methods for partial differential equations and scientific computing, with a focus on mimetic and virtual element methods on polytopal meshes and applications to porous media flow and computational geomechanics.

EDUCATION

Stanford University

Stanford, CA

Ph.D. in Computational and Mathematical Engineering

Sep 2024 – Present

Advisor: Hamdi A. Tchelepi

Technische Universität München (TUM)

Munich, Germany

M.Sc. in Computational Science and Engineering

Oct 2021 – May 2024

Thesis: *Dynamical Low-Rank Approximation of the Vlasov–Poisson Equation* ✉

Advisor: Caroline Lasser

University of California, Berkeley

Berkeley, CA

B.A. in Applied Mathematics

Aug 2018 – May 2021

RESEARCH EXPERIENCE

Graduate Research Assistant (SUETRI-B) ✉

Stanford University

Stanford, CA

Sep 2024 – Present

- Develop numerical formulations based on Mimetic Finite Difference (MFD) and Virtual Element Methods (VEM) for polytopal meshes, with applications to reservoir simulation.
- Design and analyze adaptive discretization strategies for numerical robustness and computational efficiency on complex reservoir grids.
- Implement and validate the resulting methods in research software and large-scale simulation frameworks.

Visiting Ph.D. Student

Politecnico di Torino

Turin, Italy

Jul 2026

- Initiated an ongoing research collaboration with the Numerical Analysis and Scientific Computing Group ✉.
- Develop mixed Virtual Element Method (VEM) formulations for linear elasticity on general polyhedral meshes.
- Implement and validate the resulting framework for applications in computational geomechanics.

Scientific Assistant

Max Planck Institute for Plasma Physics

Garching (Munich), Germany

May 2023 – Sep 2023

- Investigated geometric numerical integration and reduced-order modeling methods for plasma simulations.
- Developed and evaluated neural variational integrators for structure-preserving reduced-order models.

Undergraduate Research Assistant

Yonsei University

Seoul, South Korea

Jul 2019 – Feb 2020

- Conducted research in the Computational Science and Engineering (CSE-URP) program ✉.
- Developed polygonal mesh generation algorithms based on Centroidal Voronoi Tessellations (CVT).
- Investigated upwind finite difference discretizations for convection–diffusion equations with emphasis on numerical stability and boundary-layer resolution.

PUBLICATIONS AND PREPRINTS

- [1] Y. J. Cha, O. Duran, N. Castelletto, V. A. P. Magri, and H. A. Tchelepi. “Adaptive Consistency for Mimetic Finite Differences.” *Submitted to Journal of Computational Physics*, 2026. [arXiv] [Code & Data]

CONFERENCE PRESENTATIONS

A Residual-Based Criterion for K-Orthogonal Cell Classification in Mimetic Finite Difference. World Congress on Computational Mechanics (WCCM)–ECCOMAS Congress, Munich, Germany, Jul. 2026.

A Residual-Based Criterion for K-Orthogonal Cell Classification in Mimetic Finite Difference Simulation of Diffusion Problems. Computational Methods in Water Resources (CMWR 2026), Bologna, Italy, Jun. 2026.

PROFESSIONAL EXPERIENCE

Reservoir Simulation Research Intern

Stanford, CA

TotalEnergies

Jul 2026 – Present

- Develop advanced discretization capabilities in GEOS for compositional flow.
- Advance adaptive and robust discretization strategies for large-scale flow and geomechanics simulations.

Reservoir Simulation Research Intern

Stanford, CA

TotalEnergies

Jun 2025 – Sep 2025

- Contributed to scientific software development and numerical simulation workflows for subsurface flow in GEOS.
- Designed verification and benchmark studies to assess numerical accuracy, robustness and computational performance.

Data Analytics and Business Consulting Intern

Seoul, South Korea

Ernst & Young (EY)

Aug 2022 – Oct 2022

- Conducted quantitative market analyses across 10 countries, developing data processing and visualization workflows for large-scale industry datasets
- Developed standardized and reproducible methodologies for cross-market benchmarking and comparative analysis.
- Received a return offer.

FELLOWSHIPS AND AWARDS

EDGE Fellowship, Stanford University, Vice Provost for Graduate Education (VPGE)

2024

PROFESSIONAL SERVICE AND MEMBERSHIPS

Secretary, Stanford SPE Student Chapter

2026 – Present

Member, Society of Petroleum Engineers (SPE)

2026 – Present

Mentor, Mathematics Undergraduate Student Association (MUSA), UC Berkeley

2019 – 2021